



Human-centered AI for inclusive tourism: enhancing travel planning for adults with autism spectrum disorder

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HIGHLIGHTS

- We found that, despite difficulties, autistic adults wish to plan trips autonomously.
- We propose a Human-Centered AI trip planner for mid/high-functioning ASD users.
- Step-by-step guidance helps autistic users plan trips with limited effort.
- The app empowers users to make choices while AI guides the process, ensuring balanced control.
- The app balances user guidance and decision-making control.

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ABSTRACT

Travel experiences can challenge individuals with Autism Spectrum Disorder (ASD) before embarking on a trip, because of the complexity of planning, but also during the trip, due to sensory overstimulation while visiting places. Artificial Intelligence (AI) could help overcome some obstacles. However, it might represent a barrier for people with mid-functioning autism if their needs for assistance are not considered when designing tourism applications. Moreover, most trip planners fully control the generation of solutions, undermining users' decision autonomy.

To address these challenges, we explored the development of itinerary planning technologies through a human-centered approach. We aimed to balance guiding users in their decision-making with empowering them to plan tours independently. The result is CARES, an Artificial Intelligence-driven trip planner designed for autistic adults, which we present in this article. CARES applies a collaborative approach to developing itineraries, based on the exploitation of (i) AI technologies for information filtering and interactive itinerary planning that are robust to the data scarcity characterizing the autism domain; (ii) a user interface that reduces information overload and decision fatigue in information exploration.

We tested CARES with 14 autistic adults, gathering insights to guide future design and improve the usability of Artificial Intelligence for neurodivergent users. The results indicate that our approach provides an accessible solution for enhancing the travel experiences of mid- to high-functioning autistic adults. In doing so, it contributes to more inclusive tourism.

1. Introduction

At least 78 million people have been diagnosed with Autism Spectrum Disorder (ASD) worldwide (Lord et al., 2022). ASD is a neurodiversity that affects how people perceive the world and interact with others. Being a spectrum condition, it impacts people in different ways, causing cognitive problems (low-functioning, LF, or mid-functioning,

MF) or advanced intellectual abilities (high-functioning, HF).¹ The common ground is social behavior issues (Hobson, 1993) and atypical sensory sensibility (Tavassoli et al., 2014). These features impact the tourist experience, which implies unpredictable environmental stimuli and many social interactions. Planning an itinerary involves complex decisions, such as selecting the places to visit, transfer methods, and

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¹ High, medium, low functioning correspond to levels 1,2,3 in the ASD scale according to DSM5 criteria (American Psychiatric Association, 2013).

related activities (Cena et al., 2023a). Decision-making can cause anxiety, fatigue, and exhaustion. This can induce avoidance of decisions (Luke et al., 2012; Zellner and Olds, 2016). Thus, some autistic people avoid traveling (Smith, 2015; Shaw and Coles, 2004), even though living a tourist experience might improve their quality of life. Artificial Intelligence (AI) could contribute to overcoming some obstacles to the tourist experience (Guner and Acarturk, 2020). However, people with mid-functioning autism might perceive this technology as a barrier because their needs are not considered when designing tourism applications (Ali et al., 2023). Thus, they cannot effectively and independently use them.

In the CARES project,² we set the goal of developing a tourist guide supporting autistic users in autonomously planning customized itineraries. As low-functioning ASD people rely on caregivers for their activities, we targeted people with high or medium levels of autism.

We adopted a Human-Centered AI (HCAI) approach (Shneiderman, 2022) to design and develop the application. The existing literature reports that people with autism have difficulties with planning and organization, need guidance in decision-making, and desire to control their experiences; moreover, it suggests personalizing solutions to avoid perceptual overload and respect individual aversions (negative preferences). To gather further user requirements, we conducted a pre-study by administering a qualitative interview to 17 HF or MF people with ASD recruited by the staff of the Adult Autism Center of the city of Torino.³ The main topics we extracted from their responses revealed that, despite using online tour planners, most of them were challenged by a trade-off between controlling itinerary creation to obtain detailed and personalized itineraries and receiving guidance to face the difficulties of information exploration and organization. We thus focused on (i) enhancing user control over the selection of places to visit through a simple interaction style and (ii) reducing information overload during data exploration and itinerary planning. The result is the CARES app.

Different from tour planners that leave users alone in Point of Interest selection (e.g., Google My Maps⁴) or propose built-in itineraries (Sánchez and Bellogín, 2022), CARES targets autistic users' needs as follows:

- To take preferences and aversions into account and support informed decision making, it suggests candidate Points of Interest by simultaneously considering (i) the user's positive and negative preferences for categories of Points of Interest, and (ii) the available data about places that can be presented in their preview during exploration.
- To prevent information overload, CARES also supports a step-by-step design of the trip plan, executing real-time consistency checks and itinerary optimization to propose Points of Interest that fit the spatial and temporal constraints of the itinerary under construction. This is designed to help users plan trips in manageable steps, empowering them to make decisions at their own pace without feeling overwhelmed.
- To simplify the interaction style, the application only presents closed questions (Luke et al., 2012). Specifically, it enables the user to inspect one candidate Point of Interest at a time, progressively exploring other options to decide whether to add them to the itinerary or ask for further suggestions.

The work described in this paper focuses on the human-AI interaction aspects regarding tour planning. Specifically, we conducted a user study to answer the following research questions:

- RQ1: To what extent do step-by-step guidance and personalization features implemented in CARES support trip planning?

- RQ2: Does CARES provide accessible trip planning for autistic individuals with mid-to-high functioning abilities?
- RQ3: What impact does CARES have on trip planning compared to a conventional planning system?

We tested CARES in a user study with 14 people with autism (7 HF, 7 MF). As a use case, we selected the Langhe, a rural area with small cities located in Piedmont, Italy. It is suitable for organizing daily trips from Torino and can be visited by car or on a walking itinerary. The results of this test revealed that the system enables users to actively participate in decision-making while mitigating cognitive overload through step-by-step guided interactions.

This work shows that, by leveraging human-centered approaches, intelligent systems such as tour planners can address functional requirements and enhance the user experience, making technology accessible and supportive for individuals with diverse levels of autism. Our contributions are:

- An intelligent planning system that empowers users to design itineraries through guided interaction, leaving them in control of decision-making.
- An accessible tourist support system that can be used by people with high and medium levels of autism to plan trips autonomously.
- An empirical study involving high- and mid-functioning ASD individuals in a real-use scenario to assess the solution's accessibility and usefulness.

CARES employs content-based filtering, constraint satisfaction, and route optimization to manage the interaction with the user and the itinerary construction because these techniques can be used in severe data scarcity domains. Indeed, the low representation of autistic users (e.g., about 2.21 % in the U.S. (Dietz et al., 2020)) and their tendency to limit traveling, or repeat the same tour experiences, strongly challenge the acquisition of datasets enabling the adoption of more recent technologies, such as neural networks, which require model training. Another important point is that adopting constraint satisfaction and content-based filtering based on positive/negative preferences supports transparency in the presentation of suggestions to the user and the visualization of the user model, qualities that are particularly challenging to guarantee with black-box machine learning approaches.

In the following, Section 2 outlines the related work. Section 3 describes our user-centered AI approach. Sections 4 and 5 present the preliminary study and the UCD process we followed. Section 6 describes the CARES app. Section 7 describes the user study. Sections 8 and 9 present the results and discuss them. Sections 10 and 11 report our work's implications and limitations. Section 12 concludes the paper.

2. Background and related work

2.1. Background on decision-making in autism

Decision-making is a complex mental process, through which "one of two or more possible options or actions is actively selected to reach a desired goal" (Edwards, 1954). Planning an itinerary involves the selection of the places to visit, coupled with information on timing, transfer methods, and related activities. Thus, to design CARES, we considered the results of the works examining decision-making in people with autism.

A robust body of empirical work demonstrates that the decision-making processes of adults on the autism spectrum differ systematically from those of neurotypical individuals, and thus people with autism exhibit unique needs. Dual process theory (Thompson et al., 2009; Brosnan et al., 2016) distinguishes between an intuitive, fast, and automatic system (System 1) and a slower, more analytical and deliberate system (System 2). Research (Luke et al., 2012; Vella et al., 2018; Farmer et al., 2017) indicates that autistic individuals tend to favor more effortful, analytic processing (System 2), requiring extended deliberation

² CARES - inClusive personAlized touRist systEms autiSm.

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⁴ <https://www.google.com/intl/en/maps/about/mymaps/>.

time and step-by-step incremental decision-making, as rapid or holistic decision-making (System 1) often leads to fatigue, anxiety, and avoidance.

Luke et al. (2012) examined “real-life” choices made by adults with and without ASD and reported that autistic participants experienced greater difficulty when decisions had to be made quickly, required a departure from routine, or involved social negotiation. The authors observed that such situations provoked fatigue, heightened stress, and anxiety, leading many participants to avoid the decision altogether. Thus, people with autism need to reduce information overload and require some cues to focus attention on important aspects.

Vella et al. (2018) used a standardized decision-making questionnaire (rather than self-report) and found that, although autistic adults often reached more logically consistent outcomes, they required considerably longer deliberation times and gathered more information before committing to a choice. This implies that people with autism need extended processing time. This pattern was interpreted as evidence of more effortful, analytic processing. Farmer et al. (2017) arrived at a similar conclusion, showing that autistic decision-makers tend toward consistent, rational selections—a style that can be advantageous for tasks with well-defined rules (e.g., mathematics) but may be detrimental when rapid integration of social and contextual cues is essential, such as planning a trip. This implies that people with autism need a predictable interaction flow. Farmer et al. (2017) observed that autistic adults often prefer to make a series of small, verifiable decisions rather than a large, all-or-nothing choice. Thus, they need support for incremental commitment.

These findings suggest three design guidelines for any decision-support system intended for autistic adults:

- Minimization of irrelevant information - to reduce cognitive overload (Vella et al., 2018).
- Clear visual or textual presentation of the decision-relevant attributes - to focus attention on analytical aspects while shielding users from extraneous details (Vella et al., 2018).
- Provision of extended processing time and explicit, closed-question formats - to accommodate slower deliberation (Luke et al., 2012).
- Offering a step-by-step interaction (Farmer et al., 2017).

The CARES system can be framed within the dual process theory of decision-making (Thompson et al., 2009; Brosnan et al., 2016) in autism by recognizing how it addresses the distinct cognitive styles seen in autistic adults. CARES supports the decision process of System 1 by providing personalized, stepwise guidance that allows users to break down the complex trip-planning task into manageable, verifiable decisions, preventing information overload.

2.2. ICT solutions for autism and tourism

Studies about tourism, industry practice, and government policy have traditionally assumed that tourists are neurotypical rather than neurodivergent (Jepson et al., 2024). Few studies focus on the tourist experience of people with ASD, but most deal with families with autistic children (Conde et al., 2023; Sedgley et al., 2017). Adults are understudied except for a few exceptions (Zhao et al., 2023).

Technology is considered valuable in helping people with ASD (Simm et al., 2016). It is mainly used to support social activities such as talking to others, understanding emotions, collaborating, coping with stress, improving people's autonomy at home, and shopping activities. However, few technologies support people with autism in tourism. Most services are simple informative websites, like Autistic Globetrotting⁵ and Toerisme voor Autisme.⁶ They show how to pack luggage and review

tourist attractions, describing what visitors can expect there. Other attempts are tourist websites accessible to people with ASD, e.g., (Dattolo et al., 2016).

Very few works present solutions supporting people with ASD in managing the tourist experience. Varriale et al. (2022) focuses on how digital technologies can improve the inclusiveness of museums. This is an exploratory and qualitative study on the design of an app for children with autism. The app provides a digital visual agenda to support children and their families when visiting an archaeological museum.

Cena et al. (2023b) presents Piuma, a map-based mobile app useful for finding relevant places in a city. Piuma exploits places' sensory features to suggest locations suitable for users' idiosyncratic needs. It has been tested with people with high-functioning autism, and it is not suitable for people with lower cognitive capabilities due to the complexity of the interaction.

Rezae et al. (2021) propose a mobile application to support the use of public transportation. The app aims to present travel information intuitively to reduce the cognitive load involved in reading and processing service schedules. However, the planning is not personalized to users' preferences and needs. These works reveal two main gaps in the state of the art:

- Lack of a personalized, guided planning tool that supports pre-trip planning for autistic adults across the full functional spectrum.
- Lack of mixed-initiative interaction (Horvitz et al., 2005) that balances system-generated recommendations with user-controlled adjustments, a design principle shown to improve engagement and perceived control (Convertino et al., 2017; Deterding et al., 2017).

Filling these gaps is essential for translating the decision-making insights of Section 2.1 into a usable, real-world solution.

2.3. Tourism recommender systems

Most trip planners generate optimal itineraries by applying Content-based or Collaborative Filtering techniques to select the places to visit, combined with traditional routing algorithms, like Dijkstra's algorithm, to schedule visits (Gavalas et al., 2011; Kurata and Hara, 2013; Borrás et al., 2014). GPS traces have been used to mine travel itineraries for recommendations (Zheng and Xie, 2011). Moreover, location-based networks have been exploited to retrieve user-generated content (like geo-tagged photos) and check-in data for building richer user profiles and inferring widely used travel plans (Brilhante et al., 2015; Lim et al., 2018; Sánchez and Bellogín, 2022). Furthermore, Deep Neural Networks have been used to recommend places (Yu et al., 2020) and build personalized itineraries. See (Sarkar et al., 2022) for a survey.

These systems fully control the itinerary generation process, failing to involve the user in its construction, a primary goal of our work. Some tour planners let users criticize the itineraries and ask to generate different ones or remove specific items from the plans, as in (Vansteenkeweg et al., 2011). However, they fail to support equal user-system collaboration.

Most commercial trip planners, e.g., Roadtrippers,⁷ adopt a similar approach and fully control the itinerary generation. Other ones, such as Wanderlog,⁸ enable users to select the places to include in a tour. However, they lack the itinerary scheduling function.

Similar to the use cases explored in customer support management (Convertino et al., 2017) and mixed-initiative creative interfaces (Deterding et al., 2017), a tour planner can play a key role in supporting user awareness of the constructed itineraries, considering their consistency and feasibility. In this perspective, Mauro et al. (2024)'s work supports the construction of school field trips involving teachers in selecting the places and activities, but different from CARES, it does not personalize the recommendations.

⁵ <https://autisticglobetrotting.com/>.

⁶ <https://www.toerismevoorautisme.be/>.

⁷ <https://roadtrippers.com/>.

⁸ <https://wanderlog.com/>.

To summarize, no existing tourism recommender system simultaneously (i) offers a mixed-initiative, user-system collaborative planning loop, (ii) tailors recommendations to the distinctive decision-making characteristics of autistic adults, and (iii) supports users with ASD across a large functional range (high- and mid-functioning).

By integrating the decision-making constraints identified in Section 2.1 with the collaborative planning principles, we aim to create an itinerary assistant that (a) reduces unnecessary information, (b) affords flexible deliberation time, (c) presents recommendations in a clear, visual format, and (d) sustains an interactive dialogue with the user throughout the planning process. This directly addresses the gaps in both the autism-focused ICT literature and the broader tourism-recommender research, providing a novel contribution that is both theoretically grounded and practically needed.

3. User-centered AI approach in CARES

The literature discussed in Section 2.1 highlights autistic people's need to be supported in decision-making without being overwhelmed by information. However, we aimed to enhance their autonomy in trip planning. To find a trade-off between these contrasting requirements, during the design of the CARES app, we drew inspiration from the principles promoted by Shneiderman (2022) in his definition of Human-Centered Artificial Intelligence (HCAI). Shneiderman proposes to fine-tune the offer of full automation versus full control according to people's needs to avoid the dangers of excessive human or computer control. We took this principle as a building block of our work, combined with a user-centered design approach (Abbras et al., 2004; Mao et al., 2005) to elicit users' interaction requirements.

In the User-Centered Design (UCD) (Gould and Lewis, 1985), end-users influence how a design takes shape. We involved autistic people from the beginning of the CARES project to gain user requirements and assess the solution's acceptability and support for the user:

- i) In the first phase of the project, we used qualitative interviews to collect users' needs and understand their traveling issues, patterns of traveling, fears, and how they used technology. See Section 4.
- ii) Later, we applied the UCD process based on iterative cycles of design-evaluation-redesign (Section 5). This methodology involves end users in technology-based services design and is increasingly employed in the technology for autism (Maun et al., 2021; Good et al., 2016).
- iii) In the last phase of the project, we employed HCI techniques and methods to evaluate CARES's usability and trip planning support (Section 7). We asked autistic people to use CARES with a task to test the application in the context of real activities. Moreover, we applied observation and inquiry techniques to collect data about people's interactions with the app and their evaluations of the system.

4. Preliminary study

To gather requirements for the CARES app, we interviewed 17 people through qualitative interviews.

Subjects. The participants were autistic adults recruited by the staff of the Adult Autism Center. There were 6 females, 11 males, 0 non-binary, 0 not declared, with an average age of 31 (Standard Deviation = 8.36). According to DSM5 criteria (American Psychiatric Association, 2013), which we used to classify the level of autism, 13 people were HF and 4 MF.

Interview protocol. For the interviews, we employed a semi-structured approach to explore participants' travel experiences and needs. Questions were organized around four thematic areas identified by the psychologists of the Adult Autism Center: participants' interest in traveling, the approaches and strategies they use for trip planning, their

reliance on technological tools and applications before departure and while traveling, and their vision of an ideal travel experience.

Procedure. Participants were invited to the Adult Autism Center for in-person interviews; however, online interviews via video conferencing were offered as an alternative for those who preferred remote participation. All interviews were conducted by a clinical psychologist with specialized training and experience in autism spectrum disorders. Each interview session lasted approximately 1 h and was audio-recorded with participant consent. All recordings were subsequently transcribed *verbatim* to ensure accurate data capture.

Data analysis. The transcribed interviews were analyzed using qualitative content analysis (Hsieh and Shannon, 2005). Two researchers and the psychologist independently reviewed the transcripts to identify and categorize participants' responses according to predefined categories aligned with the research objectives: travel willingness, planning modalities, technology use, and ideal trip characteristics. The researchers and the psychologist met to compare findings, resolve discrepancies through discussion, and achieve consensus on categorization. Representative quotes were extracted to illustrate key findings and user needs that would inform the app design.

Results. In the following, we report and describe the main findings from the interviews grouped by the questions we asked. We translated participants' answers from Italian.

Do you like traveling? Participants like traveling (MF1 and MF3 said explicitly: "I really like traveling"). However, most HF subjects suggested that the journey should last a few days to be pleasant. For example, HF5 answered: "Yes, I like traveling. So much. But only for short periods. Not more because then it becomes hard." In other cases, the question generated confusion and doubt. For instance, MF1 declared: "Mmh... I don't understand, you know? Because I'm afraid." Moreover, some participants said that they would like to travel much more than they have. In this regard, HF4 stated: "Yes, but I can't do it as often as I would like. Why? Because I make a tremendous effort to organize".

How do you plan your trip? MF participants cannot plan a trip by themselves; they need the help of their family for the organization. Thus, they often participate in already organized trips or are accompanied by autism-specialized staff. MF2 said: "I don't know how to organize myself, maybe in Italy I could manage it, but abroad I'd need the help of my family members".

HF participants wish to plan a trip autonomously but seem aware of the difficulties they might face. HF2 remarked: "It bothers me even though I can solve any quantum physics differential equation... I don't know how to organize the trip." Usually, HF participants use the web to choose places and how to reach them (i.e., means of transportation), monuments, or what to visit once there. In the planning phase, they frequently make a list as HF8 explained: "I always make a list of things to see in the cities."

Do you use any technological tools before or during the trip? All the participants answered this question positively. Both MF and HF people use mobile phones during the trip. Specifically, HF participants use them to reach the places they want to go to. They use the web to find information, websites like Booking⁹ to book services, websites like Tripadvisor¹⁰ to choose where to eat. Moreover, they use parking apps to find a spot if they have a car. HF5 reported: "I always use my mobile phone. I don't like to use a printed map or ask people around... in fact, to always avoid this, I always bring my power bank so I can charge it". MF people rely on family members and friends to help them use booking and navigation tools, as they cannot use these systems independently. For example, MF2 said, "I need the help of my dad to plan a trip using Booking".

⁹ <https://www.booking.com>.

¹⁰ <https://www.tripadvisor.it>.

Describe how your ideal trip should be. The key for all participants is that the tour is well-organized in advance. Most of them consider planning as a way to control as many variables as possible. See the following answers. HF8: “I make a program to be respected so that everything works well”. MF4: “When I travel I need to know everything beforehand”. HF5: “Before leaving I must have everything booked, what to see, and also where to eat lunch and dinner for the duration of the trip”. HF12: “I just have one word for you: it must be organized!”. MF3: “The ideal would be to have a guide who organizes the trip or just what to do in the city but especially to do all the possible experiences that the city offers”.

User requirements. We first summarize the requirements gathered from the interviews conducted during the preliminary study.

- *Need for organization.* HF and MF participants like traveling. However, organization and planning are crucial to enjoying the travel experience, and one of the main difficulties is the travel planning phase. Participants feel stressed when planning the trip (where to go, how to get there, what to visit, etc.). Thus, they tend to delegate to someone else or give up the trip. Sometimes, the family organizes the trip for them (especially MF people). This requirement confirms the literature findings that people with autism face challenges with planning and organization, especially for tasks like traveling (Simm et al., 2016; Cromjongh et al., 2024).
- *Need for guidance in the decision.* Especially, MF participants declared that they need help in making decisions. This is in line with the literature on decision-making and autism (Luke et al., 2012).
- *Need for control.* People with ASD would like to know in advance what to do and what to expect from the experience. This confirms the findings in the literature (Simm et al., 2016; Gomot and Wicker, 2012; Rapp et al., 2018). A detailed plan could address this need.
- *Need for short trips.* People with ASD in our interviews expressed interest in traveling, but they had a preference for limited time periods. Planning trips as half-day or full-day excursions could meet this specific requirement.

Moreover, the following requirements are retrieved from the literature:

- *Need for personalized support.* Since autistic people are heterogeneous in needs and features, they need customized solutions (Farr, 2010; Cañete and Peralta, 2023; Bondioli et al., 2017; Petty et al., 2021). This is particularly true for tasks such as traveling, where people might benefit from personalized support (Cromjongh et al., 2024).
- *Respect people's aversions.* Individuals with autism spectrum disorders often experience difficulties with anger and emotion regulation. According to the literature (Samson et al., 2015; Quek et al., 2012; Ho et al., 2012), when they come across things they dislike, they can become annoyed, but also angry. Thus, their preferences and aversions must be considered in the recommendations.
- *Avoid perceptual overload.* Individuals with ASD require higher perceptual loads to ignore distractors (Remington et al., 2009; Hill and Frith, 2003). Thus, the interaction with the app must be carefully designed, avoiding unnecessary details and too much information.

5. UCD process

We followed the UCD process based on iterative cycles of design-evaluation-redesign (Gould and Lewis, 1985).

We conducted two cycles of design, evaluation, and redesign of the prototype with autistic users. Specifically, we conducted a two-stage usability testing with a proof-of-concept prototype, involving two high-functioning (HF) and two medium-functioning (MF) participants from the pre-study cohort. Usability testing, as a core user-centered evaluation technique, allowed for direct observation of user interactions and identification of usability bottlenecks, such as interface complexity and interaction inefficiencies. Feedback from the first sessions led to evidence-based refinements: simplifying the user interface, shortening

Table 1

Requirements from the interviews and the literature, and corresponding functionality provided by CARES.

User requirements	Source	CARES' functionality
Need for organization	Our interviews, literature (e.g., Simm et al., 2016; Cromjongh et al., 2024)	A support to create detailed plans
Need for guidance in the decision	Our interviews, literature (e.g., Luke et al., 2012)	A step-by-step interaction with suggestions of what to do
Need for control	Our interviews, literature (e.g., Simm et al., 2016; Gomot and Wicker, 2012; Rapp et al., 2018)	The user explicitly controls the itinerary generation; (s)he can see all options if desired
Need for short trips	Our interviews	The CARES app allows to plan half- or full-day trips
Need for personalized support	Literature (e.g., Farr, 2010; Cañete and Peralta, 2023; Bondioli et al., 2017; Petty et al., 2021; Cromjongh et al., 2024)	The support is personalized according to a model of the user
Respect users' aversions	Literature (e.g., Samson et al., 2015; Quek et al., 2012; Ho et al., 2012)	The recommendation algorithm considers also aversions
Avoid perceptual overload	Literature (e.g., Remington et al., 2009; Hill and Frith, 2003)	The CARES app is accessible to different levels of autism

textual content, employing more accessible terminology (e.g., replacing “Points of Interest” with “Places”), and increasing navigational support by adding additional back buttons, enhancements shown to reduce cognitive load and improve task completion rates.

Given this feedback, we revised the prototype app and tested it with users. In this second iteration, MF participants requested an increased use of visual aids (i.e., more images of suggested places) and a sequential presentation of candidate destinations during itinerary planning, reflecting a need for reduced information density and improved information chunking. HF participants emphasized the importance of system personalization, advocating for dynamic modification of user preferences and aversions within their profiles. Section 6 describes the final system used for the user study.

6. The CARES app

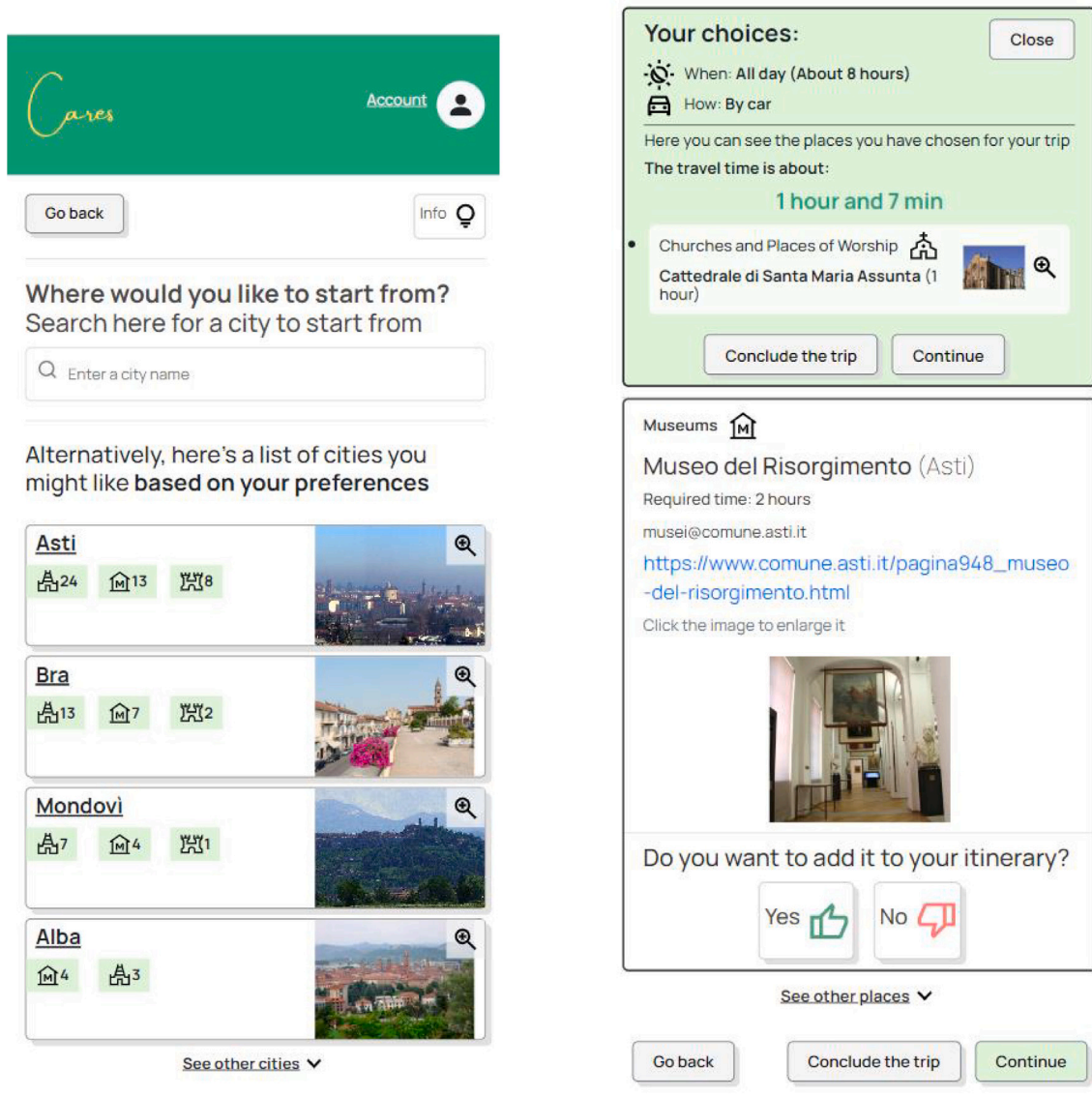
According to the requirements gathered in the preliminary study (summarized in Table 1) and during the UDC process, we designed the CARES app¹¹ with the following features in our minds:

- Supporting the decisions about planning short trips.
- Providing personalized suggestions that consider both user preferences and aversions.
- Supporting users step-by-step in the incremental design of tour plans.
- Putting users in control of the planning process.
- Proposing closed questions to simplify the interaction with the system.

6.1. Interaction with the user

CARES handles the interaction with the user in a pipeline. First, the application elicits the user's preferences (Section 6.1.1). Then, it acquires the main temporal and spatial constraints of the tour (Section 6.1.2). Then, it collaborates with the user on the trip design, suggesting Points of Interest that match the user's preferences and the tour constraints (Section 6.1.3).

¹¹ The CARES app is available at the following link: <https://cares.planner.ontomap.eu/>.



(a) Selection of the starting city of the trip.

(b) Points of Interest suggestion.

Fig. 1. Itinerary creation with the CARES application.

6.1.1. Acquisition of the user's preferences

To avoid suggesting places that might cause anxiety and stress in people with ASD (Samson et al., 2015; Quek et al., 2012; Ho et al., 2012), CARES collects the user's preferences and aversions during the registration phase.

First, it proposes a list of Point of Interest categories (e.g., castles, museums, etc.) and asks the user to select the preferred ones (positive preferences). Next, the system asks the user to identify, among the remaining categories, those representing aversions, i.e., negative preferences ("What would you **not** like to visit?"). The categories that have not been selected in the previous steps are classified as neutral valence preferences.

6.1.2. Acquisition of the spatial and temporal constraints of the tour

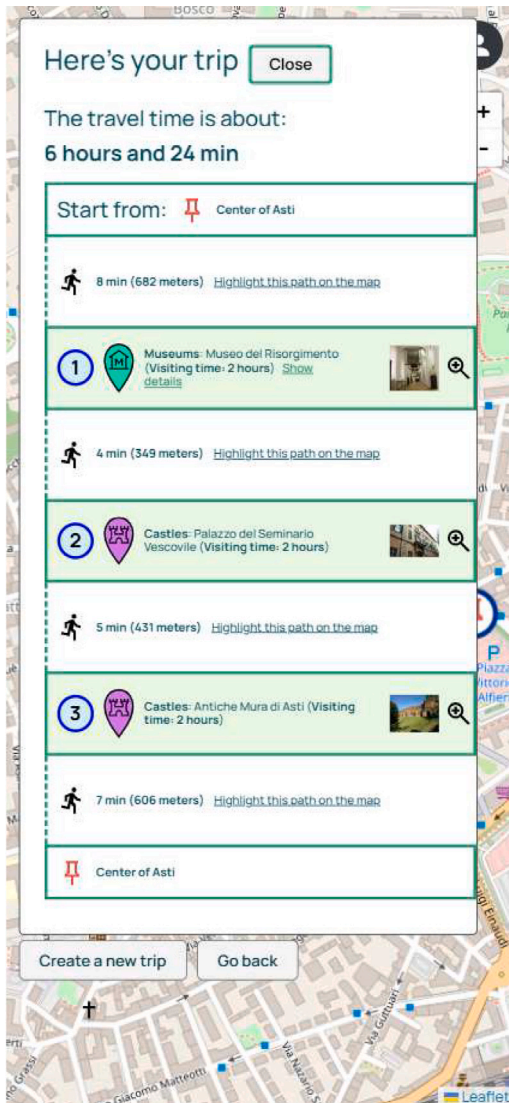
To acquire the tour constraints, we designed a simple interaction, avoiding the administration of detailed forms. First, CARES asks the user to choose between a half-day (about 5 hours) and a full-day trip (about 9 hours) and to specify whether it will be a walking or driving trip.

This last information is needed to estimate transfer times between places during itinerary scheduling.

After that, CARES guides the user in selecting the city to start the tour. This selection can be done through a classical search form, or by inspecting a closed list of recommended cities in the Langhe area which have the highest number of Points of Interest (PoIs) that belong to the user's preferred categories (Fig. 1(a)). In the recommendation list, the system supports decision-making by showing a set of icons beneath each city to summarize the number of PoIs that belong to its most prevalent categories and match the user's positive preferences.

6.1.3. Itinerary Design

The system supports trip creation by providing information about the options that match the tour constraints and that the user might like, and giving the user control over the selection of places to include. This phase of interaction is cyclic. As shown in Fig. 1(b), the app recommends a single Point of Interest at a time to prevent information overload. However, to comply with different interaction preferences, it enables the user to modify the visualization style, expanding the suggestion list by clicking



(a) Visualization of the final itinerary.



(b) Visualization of the itinerary on the map.

Fig. 2. The user interface for the visualization of the trip.

the button “See other places”. Notice that the system shows the itinerary duration (“The travel time is about:”) and PoIs’ expected visiting time (“Required time:”) to keep the user aware of the progress in the trip design and support informed decision making.

The user can add or skip each recommended Point of Interest (“Do you want to add it to your itinerary?”). Meanwhile, the app optimizes the partial itinerary and updates its duration. When the available time for the trip is exhausted or the user presses the button “Conclude the trip”, the app displays the optimized itinerary in detail (Fig. 2(a) and on a map Fig. 2(b)). Both views show the categories of PoIs as icons. Moreover, by clicking “Show details” in the detailed view, the user can visualize the data and available crowdsourced reviews of the selected Points of Interest. Furthermore, by clicking “Highlight this route on the map”, the user can view the corresponding piece of the itinerary, highlighted in red.

During the cycle, the app suggests places that best meet the following criteria that might influence the user experience (see Section 6.2 for algorithmic details):

- They can be added to the itinerary without violating its geographical and temporal constraints.

- They maximize the utility of the resulting itinerary in terms of (i) user preferences, (ii) available information about the PoIs, (iii) distance from the other places in the itinerary, and (iv) variety of the resulting itinerary. The first two criteria are relevant to suggest places that the user can appreciate and that can be reasonably previewed before selecting them (geographical information sources sometimes provide poor data about places). The third and fourth criteria support the creation of harmonic itineraries.

Overall, we designed the user interface to conform to the Web Content Accessibility Guidelines (WCAG) 2.1, ensuring robust support for users with diverse perceptual and cognitive abilities. For instance, as shown in Figs. 1 and 2, we used high-contrast color schemes to enhance visual discriminability, minimalist layouts to reduce cognitive load, and clear navigation structures.

6.1.4. Data about cities and points of interest

The system retrieves data about cities and their Points of Interest using the OpenStreetmap API.¹² The app retrieves from OpenStreetMap

¹² <https://www.openstreetmap.org>.

and Wikidata the names of the PoIs, their contacts, images, and so forth. In contrast, expected visiting times are not available in public information sources. Thus, we defined them at the system set-up time for each category of PoIs through the agreement of three researchers. For example, the estimated visiting time for the category “Museum” is 2 hours.

6.2. Points of interest recommendation algorithm

Given the city C and the selected transportation means, the app manages the itinerary construction incrementally. As previously described, the user advances from iteration I_n to I_{n+1} by adding one of the Points of Interest suggested by the system. At each iteration, I_n is an optimized partial itinerary starting from the center of C . In a driving trip, I_n ends at the last selected PoI. Differently, we assume that walking trips are circular, i.e., the start and end points of the itinerary coincide. At each iteration I_n , the system first identifies a set of feasible PoIs that can be included in I_n without violating the temporal constraints of the trip (Section 6.2.1). Then, it begins to recommend these items, one at a time, trying to maximize the utility of the resulting itinerary for the user (Section 6.2.2).

6.2.1. Identification of feasible points of interest

Let's denote the last Point of Interest in a partial itinerary I_n as $p_{current}$. The system first identifies a set $P = \{p_1, \dots, p_m\}$ of PoIs in proximity to $p_{current}$ by spatial search on the OpenStreetMap platform. The search is bounded by a circle of radius r having $p_{current}$'s geographic coordinates as its center. The radius varies depending on whether the user has chosen a walking trip (500 m) or a driving one (5 km).

Then, the system removes from P the items whose inclusion in the itinerary would exceed the predetermined time constraint T_c (5 or 9 hours). The filtering criterion is based on the following heuristic function:

$$d + \frac{airDistance(p_{current}, p_i)}{speed} + visit(p_i) \leq T_c + \epsilon$$

- d is the total duration of I_n ;
- The fraction estimates the time to move from $p_{current}$ to p_i . Specifically, $airDistance(p_{current}, p_i)$ is the geodetic distance between $p_{current}$ and p_i , expressed in kilometers. Moreover, $speed$ denotes the user's estimated travel speed, which is set to 3 km/h for pedestrian travel and 50 km/h for vehicular travel;
- $visit(p_i)$ is the estimated time to visit p_i .
- ϵ is a tolerance margin, conventionally set at 15 min, to support a flexible interpretation of half-day and full-day trips.

6.2.2. Recommendation of the next point of interest for the itinerary

As described in Section 6.1.3, we evaluate the utility of adding a feasible Point of Interest $p_i \in P$ to an itinerary I_n , by considering p_i 's distance from the last place of I_n , whether p_i meets the user's preferences, how much information about p_i is available to support decision-making, and the variety of the resulting itinerary.

In CARES, the user profile stores the user's preferences for the categories of PoIs. A user u 's profile is a vector $\vec{u} = [cat_1, \dots, cat_n]$ where cat_i is a category of Points of Interest and:

$$\vec{u}[cat_i] = \begin{cases} 1 & \text{if the user has expressed a positive preference for } cat; \\ 0 & \text{if (s)he has expressed a negative preference for } cat; \\ 0.5 & \text{otherwise (neutral preference).} \end{cases}$$

The recommendation process starts from the set $P = \{p_1, \dots, p_m\}$ of feasible Points of Interest computed in the previous phase. For each $p_i \in P$, the system computes the score ($score_i$) of the extended itinerary I_{n+1_i} obtained by adding p_i to I_n . Then, it ranks $\{I_{n+1_1}, \dots, I_{n+1_m}\}$ in decreasing order of score and proposes Points of Interest from the highest-ranked itinerary downwards.

Let us consider the user's profile ($\vec{u}$), the last PoI of the itinerary ($p_{current}$), and p_i 's category (cat). The score of I_{n+1_i} , denoted as $score_i$, is defined as:

$$score_i = mean(\vec{u}[cat], 1 - d, M, ild(I_{n+1_i}))$$

- $mean$ is the arithmetic mean and $\vec{u}[cat]$ is u 's preference for cat .
- d is the Euclidean distance between $p_{current}$ and p_i , normalized to the search radius r to ensure the result is in $[0,1]$.
- M is an indicator of the amount of metadata available in OpenStreetMap: $M = 0.5$ if any information about the place, such as a description, link to the website, or contact details, is available. If the Point of Interest also has a Wikidata tag, $M = 1$.
- $ild(I_{n+1_i})$ is the intra-list diversity of the extended itinerary. We compute diversity as the complement of its intra-list similarity:

$$ild(I_{n+1_i}) = 1 - ils(I_{n+1_i})$$

$$ils(I_{n+1_i}) = \frac{\sum_{p_i \in I_{n+1_i}} \sum_{p_j \in I_{n+1_i}, i \neq j} sim(p_i, p_j)}{|I_{n+1_i}|(|I_{n+1_i}| - 1)}$$

$$sim(p_i, p_j) = 1 \text{ if } p_i \text{ and } p_j \text{ belong to the same category, } 0 \text{ otherwise.}$$

6.3. Itinerary optimization

To optimize an itinerary I and compute its duration, CARES employs a routing service that returns the fastest path including the selected Points of Interest. For this computation, the service uses I 's start and end points, the geographical coordinates of its PoIs, their estimated visit times, and the selected transportation means.

The architecture of the routing service combines two technologies: it uses VROOM¹³ (Vehicle Routing Open-source Optimization Machine) as an optimizer. Moreover, it employs OSRM¹⁴ (Open Source Routing Machine) as a routing machine. For brevity, we omit the details of the routing service.

7. User study

To address our research questions (see Section 1), we conducted a user study focused on the scenario of people with ASD planning a trip in the Langhe region. The participants were patients of the Adult Autism Center. Our experiment has been approved by our University's Ethics Committee (Protocol Number: 0562309).

7.1. Method

We compared the CARES app with Google My Maps. We selected this service as a baseline because it shares several features with our application, and the participants were familiar with it. Even though this choice might introduce a familiarity bias, we point out that My Maps only supports searching for PoIs. As it does not incrementally suggest them, there is a clear distinction in the type of functionality the two systems offer.

For the user test, we adopted a within-subjects approach. Each participant interacted with both applications in a task experiment, which consisted of designing a trip using the applications.

7.1.1. Measures and Materials

To answer our research questions, we conducted objective and subjective evaluations of the two systems selected for the experiment. Fig. 3 shows our research framework graphically, considering our research questions. In our experiments, CARES and Google My Maps conditions are the independent variables. The figure also shows how the Participants' engagement in itinerary design and the Itinerary design support are specified into simpler evaluation dimensions.

¹³ <http://vroom-project.org/>.

¹⁴ <https://project-osrm.org/>.

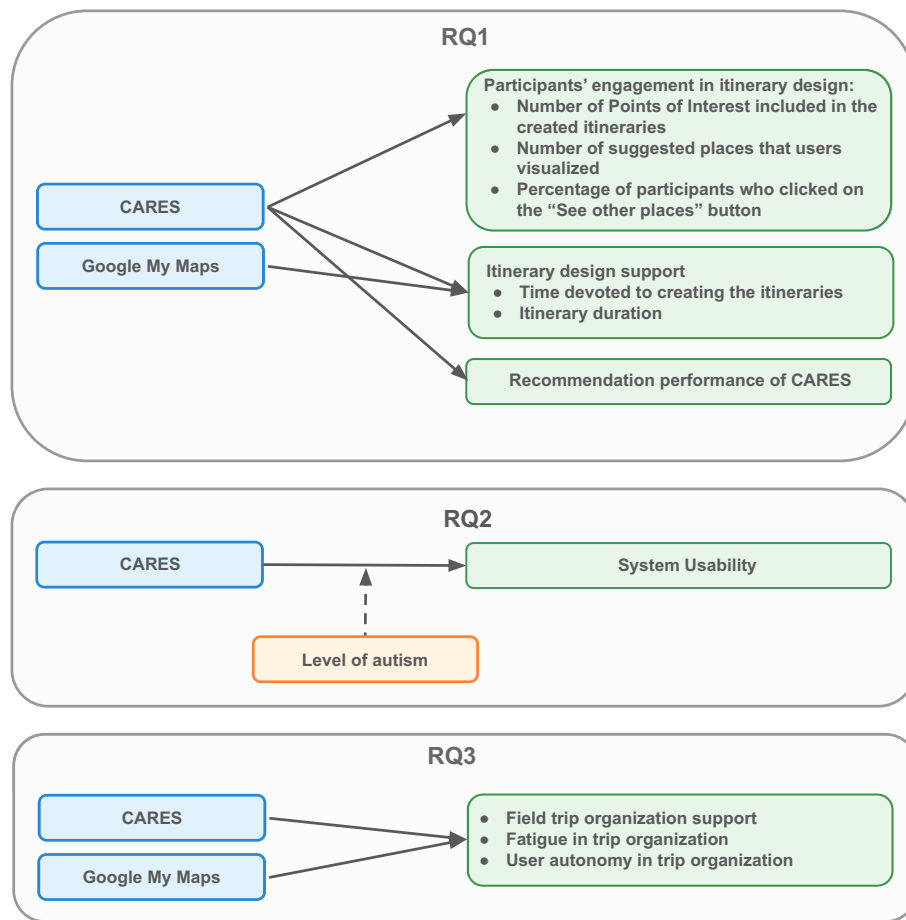


Fig. 3. Research framework representing the relationships between independent, moderating, and dependent variables, according to our research questions. We use the following color code: blue for independent variables, orange for moderating ones, and green for dependent variables. The dashed arrow links the moderating variable to the affected dependency.

Objective measures.

The objective evaluation concerns our research questions RQ1 and RQ2. The analysis is based on participants' logged actions when interacting with CARES, and a summary of their behavior with Google My Maps, annotated by the psychologist while she conducted the tests. For this evaluation, we considered the following dimensions:

- **Participants' engagement in itinerary design**, intended as the amount of work and interaction with the systems during task execution. We based this evaluation on the following data:
 - On both systems, we analyzed the **number of Points of Interest** included in the created itineraries. This indicates how articulated itineraries are.
 - On CARES, we also analyzed the following data to assess participants' level of information exploration during itinerary design:
 - * The **number of suggested places that users visualized** while creating the itineraries.
 - * **Percentage of participants who clicked on the "See other places"** button to visualize more places at each itinerary design step.
- **Itinerary design support**, evaluated as follows:
 - The **time devoted to creating the itineraries** on the two systems.
 - On CARES, we also analyzed the **itinerary duration**, considering that participants were instructed to try to fill the available time of the tours. Computing this measure on Google My Maps does not make sense because that system does not support scheduling or itinerary optimization.

- **Recommendation performance of CARES.** As Google My Maps only filters places based on the search query, this measure is irrelevant to that system. To estimate participants' acceptance of CARES' suggestions, and thus the level of personalization it offers, we computed the **percentage of accepted suggestions** (i.e., POIs included in the itineraries) **whose categories belong to users' positive, negative, and neutral preferences**. Where relevant to support data comparison (e.g., for itinerary durations), we distinguished the values concerning half-day and full-day tours. We remind the reader that participants were instructed to choose the same type of tour on CARES and Google My Maps.

Subjective measures.

The subjective evaluation concerns RQ2 and RQ3.

- We evaluated the usability of the CARES app by utilizing an Italian-translated version of the System Usability Scale (SUS) (Brooke, 1996). The SUS comprises 10 statements to be rated on a scale of [1, 5] corresponding to {Strongly disagree, Disagree, Neither agree nor disagree, Agree, Strongly agree}. See Table 5. We decided to waive the assessment of Google My Maps' usability to minimize cognitive and attentional demands on participants with ASD, who often experience stress and cognitive fatigue when faced with multiple evaluation instruments.
- At the end of the experiment, participants answered a structured interview conducted by the psychologist of the Adult Autism Center to measure the impact of trip planning compared to traditional supports. The interview contains four open questions to compare CARES

with My Maps in planning a trip, focusing on field trip organization support, fatigue in trip organization, and user autonomy in trip organization:

1. Which system would you use to organize a field trip between CARES and Google My Maps? Why?
2. With the help of CARES, would organizing a field trip be less strenuous for you than My Maps? Why?
3. Would CARES make you more autonomous in organizing a trip than My Maps? Why?
4. If you could always use a system like CARES, would you travel more? Why?

We did not find any standardized questionnaire to evaluate the aspects we were interested in. Thus, we asked the Center's team to validate the questions. The psychologist reviewed the questions for relevance, clarity, and coverage. We conducted data analysis using the qualitative content analysis approach consistent with the pre-study. Two researchers and a psychologist independently coded the answers to the questionnaire, focusing on field trip organization support, fatigue in trip organization, and user autonomy in trip organization. They then resolved discrepancies through discussion and extracted representative quotes.

7.1.2. Procedure

The user study was conducted in person, with the support of the same psychologist mentioned above, who recruited the participants. The inclusion criteria were adults with high- and mid-functioning Autism (HF and MF). The exclusion criteria were minors, nonverbal, and low-functioning individuals. We did not offer financial or other incentives to participate in the study.

We implemented a within-subjects approach, handling each treatment condition as an independent variable. Every participant experienced both treatments. To minimize the impact of practice, fatigue, and result biases, we varied the order of the tasks. We refrained from imposing time limits on task execution to let participants explore the information presented and select the PoIs for the trips without feeling rushed. Specifically, we organized the user study according to the following steps:

1. The psychologist explained the study's objectives and reassured the participants that any data collected would be used solely for scientific purposes and processed anonymously in line with European legislation. Then, she asked them to review the informed consent¹⁵ and provide explicit consent to participate by signing it. The psychologist did not gather their names or personally identifiable information to preserve participants' privacy.
2. The participants filled in a questionnaire concerning demographic information (gender, age), their jobs, and how often they take trips (often, little, rarely, never).
3. Participants were instructed to perform the same tasks using the two platforms. In particular, the instruction was: "*Plan a half-day (5 hours) or full-day (9 hours) itinerary that satisfies the following constraints:*"
 - *The itinerary has to take place in the territory of Langhe, in Italy.*
 - *The itinerary has to approximately fill the entire period (half/full day).*
 - *You should plan a trip for yourself (alone).*
 - *Transportation mode is restricted to car or foot, limiting route variability."*

To support the comparison between systems, when a participant started the second task, the psychologist asked her/him to choose the same transportation means and type of trip (full/half-day) as in

the previous task. Moreover, to avoid a bias concerning the knowledge of the visited area, she asked the participant to start the trip from a different place.

4. They used the CARES app and Google My Maps in counterbalanced order.
5. After using CARES, the participants completed the SUS questionnaire to give feedback on its usability. As previously specified, we did not evaluate My Maps' usability to avoid overloading them.
6. At last, the psychologist administered the open questions to compare CARES with My Maps in planning a trip.

7.1.3. Participants

The psychologist of the Adult Autism Center could recruit 14 participants who were not involved in the pre-study because the others declined the new invitation. The participants were 2 female, 9 male, 0 non-binary, and 3 not declared. They were 20–33 years old with an average age of 24.5. According to [American Psychiatric Association \(2013\)](#), there were 7 HF and 7 MF individuals.

Most subjects (9) stated they often take trips; 3 said they do not travel frequently, and 2 of them rarely.

8. Results

8.1. Log analysis

To answer RQ1 and RQ2 we analyzed the logged participants' behaviors. Six people (5 MF and 1 HF) created half-day itineraries, and 8 (2 MF and 6 HF) created full-day ones. We remind the reader that people could choose the type of itinerary to create, but they had to be consistent across systems. Thus, these values are the same for CARES and Google My Maps. All users completed the assigned tasks by finalizing itineraries that included at least two Points of Interest on both systems.

8.1.1. Participants' engagement in itinerary design

We first compare the two systems on the **number of Points of Interest** included in the created itineraries. [Table 2](#) shows that the mean number of PoIs included in half-day itineraries was 2.5 (SD: 0.55) with CARES and 2.67 (SD: 0.52) with Google My Maps. On average, full-day itineraries included 5 (SD: 2) PoIs in the CARES condition and 4.25 (SD: 1.58) in My Maps.

While these values are similar, the participants who created full-day itineraries included a larger overall number of places when using CARES (Total number of included PoIs: 40) than with My Maps (34). This suggests that, when planning longer trips, the itinerary support provided by CARES enhances user engagement in the planning activities. Concerning CARES:

- **Number of suggested places that users visualized.** While designing an itinerary, participants viewed on average 3.83 PoIs for half-day tours and 10.13 for full-day ones. Compared with the mean number of places included in the itineraries, this shows that users performed a reasonable amount of information exploration activity.
- **Percentage of participants who clicked the "See other places" buttons.** [Table 4](#) (graphically shown in [Fig. 4](#)) shows that 35.71 % of participants clicked this button to view more Points of Interest in a single user interface page. Specifically, 57.14 % of the HF users used this function to receive more information from the system. In comparison, the percentage of HF people asking for more information was much lower (14.29 %). This finding aligns with the features of autism profiles, which explain HF people's preference for less guided interaction and greater autonomy in exploring options. HF individuals often show more independence than MF in decision-making and information processing ([Levin et al., 2019](#)). Our findings also suggest that, with individuals with higher cognitive abilities, the system tends to promote a preference for information-seeking behavior.

¹⁵ The consent and the text describing the experiment are available at the following link: <http://bit.ly/3TTksV6>.

Table 2

Descriptive statistics of half/full-day itineraries created by the participant sample using CARES and Google My Maps. Half-day itineraries are approximately 5 hours, while full-day ones are approximately 9 hours. We report the Mean values with Standard Deviation.

	CARES half day	CARES full day	Google my maps half day	Google my maps full day
Mean itinerary duration (in seconds)	12,704 (3874)	24,906 (7748)	–	–
Total number of included PoIs	15	40	16	34
Mean number of included PoIs	2.5 (0.55)	5 (2)	2.67 (0.52)	4.25 (1.58)
Total number of suggested PoIs that users visualized	23	81	–	–
Mean number of suggested PoIs that users visualized	3.83 (0.98)	10.13 (6.20)	–	–

Table 3

Time to complete the design of half/full-day itineraries using CARES and Google My Maps (whole participants' sample, MF, and HF users). Where relevant, we report the Mean values with Standard Deviation.

	CARES			Google my maps		
	Full sample	MF	HF	Full sample	MF	HF
Mean completion time of half-day itineraries in seconds	343 (123)	340 (137)	359 (–)	641 (249)	607 (262)	812 (–)
Mean completion time of full-day itineraries in seconds	564 (509)	628 (499)	543 (558)	764 (523)	789 (191)	756 (613)

Table 4

Recommendation performance of CARES computed from logged user actions. Results concern the full participants' sample and its split into HF and MF users.

	Full sample	MF	HF
Percentage of accepted suggestions that belong to users positive preferences	78.18 %	77.27 %	78.79 %
Percentage of accepted suggestions that belong to users neutral preferences	20.00 %	22.73 %	18.19 %
Percentage of accepted suggestions that belong to users negative preferences	1.82 %	0 %	3.03 %
Percentage of users who clicked the "See other places" buttons	35.71 %	14.29 %	57.14 %

8.1.2. Itinerary Design Support

- **Time devoted to creating itineraries.** Table 3 shows that participants created itineraries more quickly with CARES than with Google My Maps, with relevant duration differences in both systems. The mean completion time of the half-day itineraries was 343 s (5 min and 43 s) in CARES and 641 s (10 min and 41 s) in My Maps. For full-day schedules, it was 564 s (9 min and 24 s) in CARES and 764 s (12 min and 44 s) in My Maps. We notice a particularly high Standard Deviation concerning full-day itineraries in CARES (about 8 min). Given the small size of our dataset, this is due to the presence of an outlier HF participant who created an itinerary of about 7 hours (including 5 places to visit), in less than 4 min. According to the answers to the questionnaires, this user felt particularly at ease with the system, and this is probably the reason for being so quick. By splitting the participants' sample into MF and HF people, we notice that:
 - When interacting with CARES, HF participants took much more time than MF ones in itinerary design (see Table 4): 359 s against 340 for half-day tours, and 543 against 628 for full-day ones. This finding suggests that HF participants devoted more time to information reading and exploration than the other group.
 - On Google My Maps, the situation is more homogeneous: on average, HF participants spent 756 s to plan full-day tours, against 789 for MF ones. Only one HF participant decided to plan a half-day tour; thus, we cannot discuss the data concerning this type of itinerary.
- **Itinerary duration.** Table 2 reports the Mean and Standard Deviation of the duration of itineraries created using CARES. The mean value of half-day tours is 12,704 seconds (3 hours and 31 min) with a Standard Deviation of 3874 s (1 h and 4 min). For full-day itineraries, the mean value is 24,906 seconds (6 hours and 55 min), with SD = 7748 s (2 hours and 9 min). Note that all the itineraries

created cover most of the available travel time. The exceptions are 2 half-day itineraries of about 2.5 hours and 2 full-day ones of about 3.5 hours, which affect the Standard Deviation.

8.1.3. Recommendation performance of CARES

Percentage of accepted suggestions whose categories belong to users' positive, negative, and neutral preferences. As shown in Table 4 (and graphically in Fig. 4), 78.8 % of the items included in the itineraries matched the participants' positive preferences for PoI categories. Moreover, 20 % matched their neutral preferences. Only 1.82 % of PoIs belonged to their negative preferences. The split into MF and HF participants reveals that HF users were the only ones to select places corresponding to negative preferences. We interpret this finding as a sign of higher flexibility in selecting what to visit, regardless of their mainstream preferences.

Overall, these findings suggest that CARES could satisfy the preferences stored in the user profiles in most cases.

8.2. Usability evaluation

To answer RQ2, we used the SUS questionnaire. Table 5 and Fig. 5 present the results (statements Q1-Q10). CARES achieved a SUS score of 74.82/100. According to the Sauro-Lewis grading scale (Lewis and Sauro, 2018), this score translates to a B grade. While SUS primarily measures perceived ease of use, it also provides insights into overall user satisfaction, particularly learnability (items Q4 and Q10) and usability (the remaining statements). CARES recorded a learnability score of 71.43/100, indicating that users found it relatively easy to learn how to use the application on their first attempt. Its usability score was 83.26/100, indicating good overall usability.

Participants reported high levels of function integration (Q5: M = 4.21, SD = 0.89). They expressed strong confidence in using the system (Q9: M = 4.14, SD = 0.95) and found it easy to use (Q3: M = 4.07, SD = 1.21). Moreover, the results indicate user autonomy. Users disagreed with statements suggesting they would need technical support (Q4: M = 2.36, SD = 1.95) or extensive preliminary learning (Q10: M = 1.93, SD = 1.44). This suggests CARES is intuitive and self-explanatory. The app also received positive feedback regarding its complexity levels. Users disagreed with the statements, suggesting the system was cumbersome (Q8: M = 1.86, SD = 1.29, lowest mean score) or inconsistent (Q6: M = 2.00, SD = 1.41). The fairly low scores for negative statements indicate good usability. Users expressed positive sentiments about CARES' learnability. They strongly agreed that most people would learn the system quickly (Q7: M = 4.00, SD = 0.96) and showed interest in frequently using it (Q1: M = 4.00, SD = 0.96).

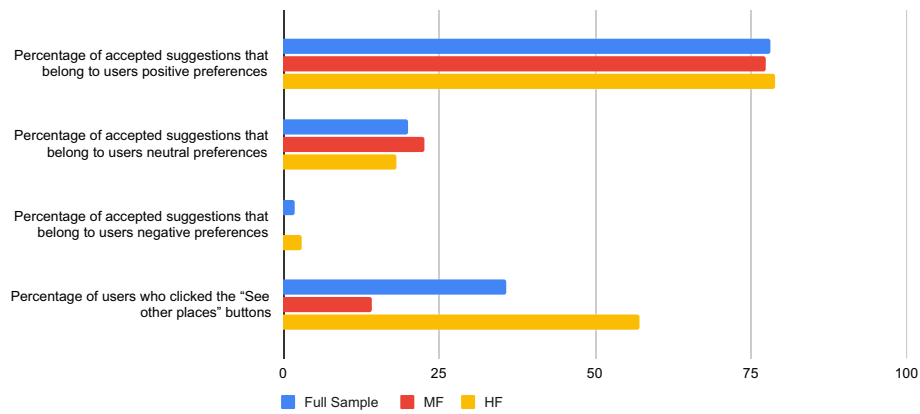


Fig. 4. Recommendation performance of CARES computed from logged user actions (whole participants' sample, MF, and HF users) in graphical format.

Table 5

Post-task SUS questionnaire results concerning the full participants' sample and its splits into HF and MF users. We report the mean value of users' responses with Standard Deviation. The best value is the minimum for Q2, Q4, Q6, Q8, and Q10, and the maximum for the other statements.

	Full sample mean(St.Dev.)	HF mean(St.Dev.)	MF mean(St.Dev.)
Q1. I think that I would like to use this system frequently.	4.00(0.96)	4.00(0.82)	4.00(1.26)
Q2. I found the system unnecessarily complex.	2.36(1.34)	2.43(1.51)	2.33(1.37)
Q3. I thought the system was easy to use.	4.07(1.21)	4.14(1.07)	3.83(1.47)
Q4. I think that I would need the support of a technical person to be able to use this system.	2.36(1.95)	1.57(1.51)	3.50(1.22)
Q5. I found the various functions in this system were well integrated.	4.21(0.89)	4.14(0.69)	4.33(1.21)
Q6. I thought there was too much inconsistency in this system.	2.00(1.41)	2.29(1.38)	1.83(1.60)
Q7. I would imagine that most people would learn to use this system very quickly.	4.00(0.96)	3.86(0.90)	4.00(1.10)
Q8. I found the system very cumbersome to use.	1.86(1.29)	1.57(0.79)	2.33(1.75)
Q9. I felt very confident using the system.	4.14(0.95)	4.00(0.82)	4.33(1.21)
Q10. I needed to learn a lot of things before I could get going with this system.	1.93(1.44)	1.71(1.51)	2.33(1.51)

8.2.1. Usability analyzed according to participants' levels of autism

Our analysis of the SUS questionnaire revealed distinct patterns and marked differences between participants with HF and MF autism concerning CARES. HF people rated the app with a SUS score of 76.43/100, indicating good overall usability. In comparison, MF users provided a slightly lower rating with an SUS score of 70.42/100.

HF users found several advantages in CARES. They reported much lower need for technical support ($M = 1.57$) than MF users ($M = 3.50$). They also found the system less cumbersome ($M = 1.57$ vs $M = 2.33$) and declared that it required less initial learning ($M = 1.71$ vs $M = 2.33$). They rated the system's ease of use slightly higher ($M = 4.14$) than MF users ($M = 3.83$). MF users showed stronger appreciation in other aspects. They reported better system integration ($M = 4.33$ vs $M = 4.14$), higher user confidence ($M = 4.33$ vs $M = 4.00$), and perceived less system inconsistency ($M = 1.83$ vs $M = 2.29$). They also indicated

slightly better learnability for most people ($M = 4.00$ vs $M = 3.86$). Both groups showed identical interest in frequent system use ($M = 4.00$).

8.3. Qualitative comparison between CARES and google my maps

To answer RQ3, we analyzed the users' answers to the open questions to compare CARES with My Maps in planning a trip (step 6 of Section 7.1.2).

12 out of 15 participants stated they preferred CARES to Google My Maps, citing several differentiating features that influenced their choice. Users particularly valued CARES' automated functionality and structured approach to trip planning. The system's ability to suggest PoIs based on user preferences was frequently mentioned as a significant advantage. Additionally, the integrated time management features for both visits and travel were highlighted as crucial elements that enhanced the planning experience. As one participant noted, "CARES is more complete and provides better information, particularly regarding visit duration and distance between locations." Several participants emphasized how this structured approach simplified the planning process, with one user stating, "It would make planning clearer and simpler because I wouldn't need to constantly seek help to plan anything." Moreover, users highlighted the system's ease of use and its ability to generate comprehensive route suggestions tailored to individual preferences: "It is easier and provides solutions with complete routes based on my preferences". Conversely, 3 participants preferred My Maps, primarily because of its familiarity. These users valued My Maps' universal applicability and previous experience with the platform, with one participant noting, "Everyone knows it and knows how to use it."

Regarding people's *autonomy*, most respondents indicated that CARES would enhance their independence in trip planning, thanks to its automated suggestions, integrated features, and reduced need to consult multiple information sources. However, while they preferred CARES for trip planning, they indicated that such a system alone would not necessarily increase travel frequency. The reason is that there are other limiting factors, such as financial constraints, personal travel preferences, and existing travel habits.

Regardless of the observed familiarity bias (involving 3 participants), these results suggest that CARES represents a promising approach to enhancing autistic people's trip-planning capabilities. The system's structured guidance and automated features align well with their needs, as evidenced by the positive feedback regarding its step-by-step approach to travel planning.

9. Discussion

Our analysis highlighted different preferences between the HF and MF participants in the user study regarding the functionality of the CARES app. HF participants found the system easier to use with

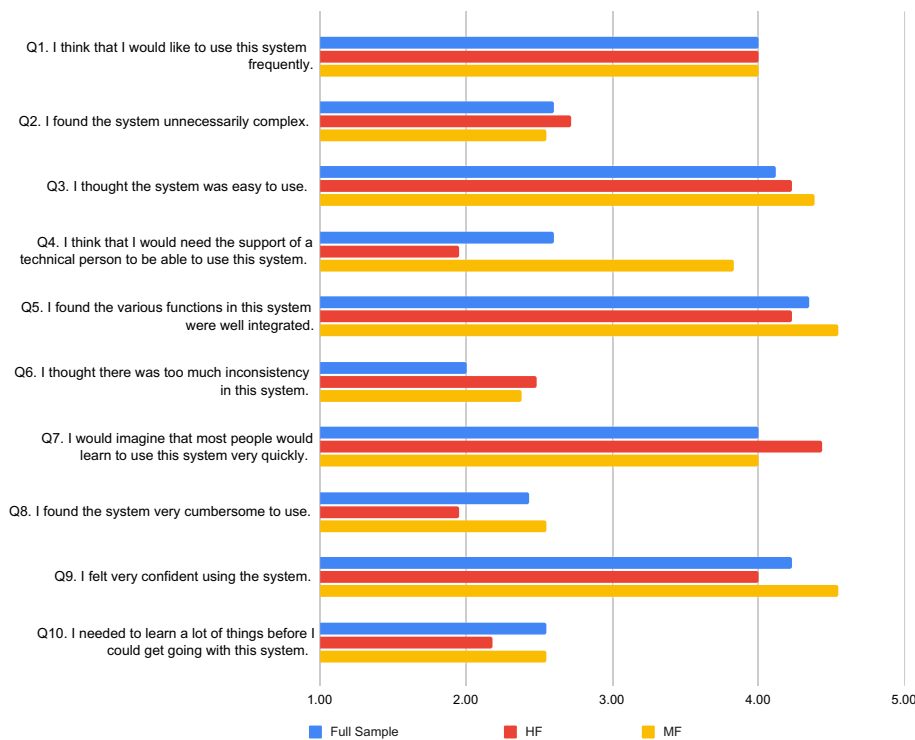


Fig. 5. Post-task SUS questionnaire results (whole participants' sample, MF, and HF users) in graphical format.

autonomy. In contrast, MF users appreciated its integrated features and showed higher confidence despite requiring technical support. Concerning our research questions:

RQ1. According to our objective and subjective findings, a guided planning system with personalized step-by-step recommendations is beneficial for autistic individuals when creating an itinerary from different points of view (*need for organization requirement*):

- It supports decision-making in itinerary design. Our objective evaluation shows that participants carried out this task more quickly with CARES than with Google My Maps. Moreover, while interacting with CARES, they created itineraries filling a good portion of the time available for the tours (half/full day). This type of support helps solve the problem of taking too long to make decisions, which often leads to inaction (Luke et al., 2012; Vella et al., 2018).
- Participants with mid-functioning autism particularly appreciated being guided by recommendations, as shown by the fact that most of them accepted all the system's suggestions. This is in line with the findings of our preliminary interviews: none of the MF participants could plan a trip autonomously, and they needed the family's help for the organization (*need for guidance requirement*). Moreover, it means that being personalized to users' preferences (according to (Farr, 2010)) and aversions (according to (Samson et al., 2015)), the recommendations were good enough to represent the participants (*need for personalized support and respect users' aversion requirements*).
- People with high-level autism appreciated having control over the recommendations and wanted to see more suggestions than the first option proposed by the system. Indeed, they spent much more time than HF participants interacting with the system, showing a longer information exploration activity. This is coherent with the *need for control requirement* (Gomot and Wicker, 2012).

RQ2. Our objective analysis shows that all participants could create nontrivial itineraries using both systems. However, the overall number of places included in full-day itineraries was higher with CARES than

with Google My Maps. We also tracked a good exploration activity on CARES. These findings support the hypothesis that people could use the system successfully. The subjective evaluation confirmed that CARES was understandable and easy to use. Moreover, HF and MF participants did not consider it cumbersome. Specifically, MF users appreciated its consistency and did not find it tiring to use the app. This may stem from the fact that we have been careful to limit the data in the user interface pages. Our goal was to avoid information overload and simplify comprehension, according to the *avoid perceptual overload requirement* (Remington et al., 2009; Hill and Frith, 2003). Thus, we can conclude that the system is accessible across different levels of autism.

RQ3. Compared to a baseline tour planner lacking recommendations and step-by-step guidance (Google My Maps), CARES was preferred by HF and MF participants, who found it less tiring (*need for guidance in the decision, and avoiding perceptual overload*). The fact that guided decision-making makes the interaction less exhausting aligns with findings in the literature, which indicate that decision-making challenges autistic individuals (Luke et al., 2012). However, even though participants appreciated our system, they declared that the actual possibility of traveling depends on additional factors beyond technology. The literature confirms this (Dempsey et al., 2021; Hamed, 2013; Freund et al., 2022).

10. Implications

10.1. Research implications

Previous research on inclusive tourism has focused on individuals with physical disabilities and those with mobility or visual impairments. Although researchers are increasingly addressing other factors, such as dementia (Innes et al., 2016) and intellectual disabilities (Gilovic, 2019), autism remains largely overlooked. Our work extends the research on inclusive tourism by highlighting that autistic individuals can benefit from an AI trip planner designed to meet their needs and requirements. Previous research stressed that people with ASD are less

Table 6
Design guidelines for AI-supported trip planners.

Design guideline	Rationale
Provide guided support for planning while allowing the option to take control at any time.	People with autism, especially MF, need to be supported in the decision. However, it is fundamental, especially for HF, to leave them the control.
Provide personalized support.	Each person's specific characteristics (Ricci et al., 2022) should be considered.
Use people's aversions when recommending places.	Samson et al. (2015); Quek et al. (2012); Ho et al. (2012), notice that people with ASD can become annoyed, or angry when they come across things they do not like. Thus, the recommendation algorithm must consider users' preferences and dislikes to avoid suggesting unpleasant items. Suggesting places that users don't like might ruin their experience.
Mimimize information.	People with ASD might be overwhelmed by information abundance. Reducing the amount and complexity of the data that the system provides can reduce this effect.

likely to travel because of anxiety problems related to unknown situations (Simm et al., 2016). Despite the many issues that might demotivate them from traveling, a system like CARES could encourage autistic people to engage in tours by enabling them to organize trips in advance, thus reducing the anxiety arising from novel situations (Neo and Flaherty, 2019; Hamed, 2013). This can impact quality of life because being able to travel is a beneficial factor (Zhao et al., 2023). Notice also that neurotypical people (friends, family members, or caregivers) could benefit from the recommendation capabilities of the system to plan suitable itineraries for their autistic companions, even those with low-functioning autism who are cognitively unable to use technological tools.

The findings from our study may also contribute to the HCI literature on the design of interactions for people with autism, whose idiosyncratic needs require technological tools that focus on how they make decisions. This implies providing personalized support in a step-by-step interaction modality (useful to MF people) and giving control to HF people over the recommendations. This type of interaction enables the system to avoid information overload (Koltay, 2017), presenting closed questions, as suggested in the autism literature (Luke et al., 2012).

Furthermore, tourist recommender systems traditionally rely on user preferences or context (Gavalas et al., 2014; Ricci, 2022), to elaborate content recommendations. We contribute to this research trend by showing that a recommender system that considers both users' preferences and aversions may be a good solution for people with autism. Specifically, CARES was appreciated by people with high- and mid-functioning autism.

Finally, we adopted a "design for all" approach, targeting users with different levels of autism and thus different levels of cognitive ability. Most solutions for autism target high-functioning users (Mauro et al., 2020). Instead, we involved high- and mid-functioning users in all the CARES trip planner design stages. Although effective engagement of autistic users can be challenging (especially the MF ones), research indicates that involving them in the design process of digital technologies can lead to increased self-efficacy and empowerment (Maun et al., 2021).

10.2. Practical implications

Our work also provides some *practical implications for the design of user-centered AI systems* that support people with autism in planning a tourist trip. Specifically, our user study helped us define a set of guidelines that can be followed to make such systems more accessible and useful to people with ASD. Table 6 presents them in a compact format.

The practical implications of CARES extend beyond general UX principles by contextualizing and tailoring well-established guidelines to the distinct needs of autistic adults planning tourist trips. While user control is a widely recognized design principle, CARES refines this concept by emphasizing a dynamic balance between guided support and user autonomy specifically shaped by empirical insights into autistic decision-making. This includes catering to the dual cognitive styles of autistic users who require extended processing time, step-by-step incremental decisions, and minimization of information overload to prevent fatigue and anxiety. The system does not merely apply generic user control, but adapts it to respect the variability of mid- and high-functioning autism, supporting both the need for structured guidance (preferred by mid-functioning users) and autonomy in exploration (valued by high-functioning users), thus offering a mixed-initiative interaction model uniquely responsive to the neurodivergent context.

11. Limitations and future work

Despite our efforts, this research could be affected by the following limitations that we will address in our future work.

First, the modest size of the participants' sample might affect our findings. The user study involved 14 people, but this number aligns with the small number of autistic users worldwide. Moreover, reaching them for research purposes often proves challenging given social difficulties. Future work should focus on extending the set of users across different geographical regions and cultural contexts, possibly through partnerships with other autism organizations and support groups. Furthermore, implementing a longitudinal study design could provide deeper insights into how users' needs evolve.

Another limitation of our research stems from the lack of continuity between the pre-study and user-study participants. Despite our efforts to maintain consistent participant involvement throughout the research phases, the users involved in the pre-study declined to participate in the subsequent user study. Therefore, we had to recruit a new group of participants.

Regarding the interaction model, the interactive approach to trip planning developed in CARES may present challenges for users with different levels of technological literacy. The step-by-step guidance model, suitable for autistic users, might not be optimal for all users, particularly those preferring more autonomous planning experiences. Future development should explore adaptive interface designs that automatically adjust the balance between user and system control according to users' technological proficiency and autonomy preferences.

Data source restrictions are challenging. We exploited OpenStreetMap (OSM) because it is an open-source system providing free APIs for retrieving geographical data. However, it introduces limitations due to inconsistent and sparse metadata coverage. OSM provides basic geographical information. However, it lacks crucial sensory and accessibility details relevant to autistic users, such as noise levels, crowding patterns, or sensory characteristics of locations. This limited metadata affects CARES's ability to make informed recommendations based on specific sensory and accessibility needs.

Another limitation of this study is that while CARES was evaluated using the System Usability Scale (SUS), a comparable assessment was not conducted for Google My Maps. As previously mentioned, we made this decision to minimize cognitive load. However, we aim to conduct further experiments in the future to extend our evaluations to both systems.

12. Conclusions

Our research addresses the critical need for inclusive and intelligent travel planning solutions focusing on the unique requirements of people with Autism Spectrum Disorder. By adopting a human-centered Artificial Intelligence approach, we aimed to support decision-making by maintaining users' autonomy throughout the trip-planning process. Our primary goal was to reduce the cognitive load and anxiety that autistic individuals often experience in this type of activity. We achieved

this with a personalized trip planner including a recommender system that responds to the user's preferences and needs. Our approach emphasizes user control and customization, allowing individuals to maintain agency in their decision-making while receiving appropriate guidance and support.

This research represents a step toward creating inclusive travel planning tools that acknowledge and accommodate neurodiversity. By prioritizing user control and personalization, our solution aims to bridge the gap between traditional travel planning applications and autistic travelers' needs, ultimately contributing to more accessible and less stressful travel experiences.

CRedit authorship contribution statement

Noemi Mauro: Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – review & editing, Visualization. **Fabio Ferrero:** Validation, Software, Investigation. **Liliana Ardissono:** Writing – original draft, Investigation, Formal analysis, Conceptualization, Writing – review & editing. **Federica Cena:** Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization, Writing–review & editing.

Code availability

The code for the web app can be found here: <https://github.com/noemi-mauro/CARES-app>.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data is available on request.

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